

VS

VS Sir Physics

AI-Enabled Physics Educator

NEET | JEE MAIN | JEE ADVANCED

★ PREMIUM HANDWRITTEN EDITION ★

ELECTROSTATICS

*Electric Charge • Electric Field • Coulomb's Law • Gauss's Law •
Electric Potential*



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15+ Years Experience • Allen | Akash | Physics Wallah

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50

Handwritten Pages



NEET + JEE Pattern Qs



Derivation-First Concepts



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About VS Sir

VS Sir is an experienced Physics educator with more than 15 years of teaching experience, having taught at some of India's most reputed coaching institutes – Allen, AKash, and Physics Wallah – guiding thousands of students preparing for NEET and JEE.

- Faculty experience at Allen, AKash & Physics Wallah
- NEET and JEE oriented conceptual teaching
- India's emerging AI-enabled Physics educator

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15+
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Teaching Philosophy

CONCEPT → VISUALIZATION → DERIVATION → APPLICATION → PRACTICE

AI LEARNING TIP + GOT A DOUBT?

Ask an AI tutor: "Why is $E = 0$ inside a conductor?" then check Part VIII. For anything else – WhatsApp VS Sir at +91 96024 05311 for feedback or NEET/JEE guidance.

This educational material has been independently prepared for learning and revision. Examination names and syllabus references belong to their respective organisations. Always verify the latest official syllabus and guidelines. No endorsement by NTA or any coaching organisation is claimed.

HOW TO USE THIS BOOK

- 1 Understand the concept**
Read the concept in simple language before touching any formula.
- 2 Visualize the physics**
Study the hand-drawn diagram – physics is easier when you can see it.
- 3 Learn the fundamental equation**
Every formula is boxed in green for quick eye-tracking.
- 4 Understand the derivation**
Follow each derivation step instead of memorising the result.
- 5 Solve the basic example**
Apply the idea to a simple, direct situation.
- 6 Attempt NEET-level questions**
Marked with the  NEET ESSENTIAL label.
- 7 Attempt JEE-level problems**
Marked with  JEE MAIN I  JEE ADVANCED labels.
- 8 Revise the formula**
Use the Formula Bank (Page 47) a day before your exam.

Recommended Roadmap

CONCEPT → FORMULA → APPLICATION → PRACTICE → REVISION

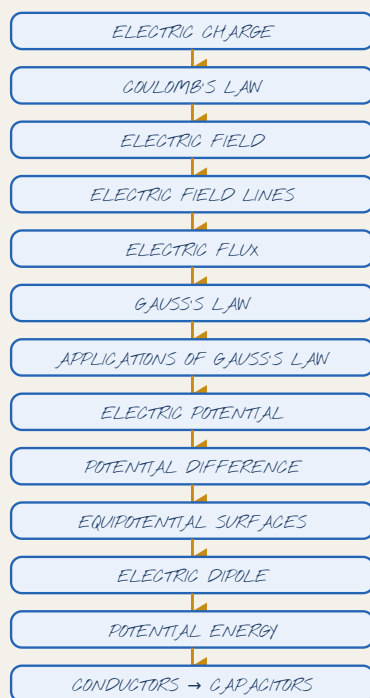
HOW THE BOXES WORK

Box	Meaning
Green box	Important formula
Pink box	Critical result / warning
Purple box	Exam TRICK
Cloud note	Reminder

BEFORE YOU BEGIN

Keep a notebook beside this book. Re-derive every boxed formula once in your own hand, attempt every  NEET PYQ PATTERN box on first sight, and revisit the Formula Bank (Page 47) a day before your exam – this simple habit outperforms passive reading by a wide margin.

ELECTROSTATICS ROADMAP



This book (Electrostatics) covers everything up to Conductors. Capacitors begin the next book in this series.

Why Electrostatics Matters

Electrostatics is the foundation of the entire Electricity & Magnetism section in NEET and JEE – Current Electricity, Capacitors, and Magnetism all build directly on the concepts of charge, field, and potential taught here.

Exam	Typical Weightage	Question Style
NEET	2–3 questions	Direct formula & concept-based
JEE Main	2–3 questions	Formula + short numerical
JEE Advanced	1–2 questions	Multi-concept, often with Gauss's law or dipole

Weightage is approximate and can vary year to year – always cross-check with the latest official syllabus and previous year analysis.

Electric Charge

1. What is Electric Charge?

Electric charge is a fundamental property of matter that causes it to experience a force when placed in an electric field, and to produce an electric field of its own. It is as basic to matter as mass.



2. Two Kinds of Charge

POSITIVE CHARGE

Charge left on glass rod after rubbing with silk. Deficiency of electrons.

NEGATIVE CHARGE

Charge left on ebonite rod after rubbing with fur. Excess of electrons.

3. Nature of Force Between Charges



- Like charges (+,+ or -,-) → repel each other
- Unlike charges (+,-) → attract each other

QUICK CHECK

Why does a comb rubbed in dry hair attract small bits of paper?

● NEET PYQ PATTERN

Q. A plastic comb rubbed with dry hair attracts small pieces of paper. This is because the comb

Ans. acquires charge by friction, and the resulting electric field attracts (polarises) the neutral paper bits.

Properties of Charge

1. Quantization of Charge

Charge on any object always exists as an integral multiple of the elementary charge e . It cannot take a fractional value.

$$Q = n e, n = 1, 2, 3 \dots$$

ELEMENTARY CHARGE

$$e = 1.6 \times 10^{-19} \text{ C (charge of a proton} = +e, \text{ of an electron} = -e)$$

2. Conservation of Charge

The total charge of an isolated system remains constant. Charge is neither created nor destroyed – it is only transferred from one body to another.

3. Additivity of Charge

Total charge of a system is the algebraic (scalar) sum of all individual charges present in it.

$$Q = q_1 + q_2 + q_3 + \dots + q_n$$

4. SI Unit & Dimension

Quantity	SI Unit	Dimension
Electric charge (Q)	coulomb (C)	[A T]

TRICK

Charge is a scalar quantity – always add charges algebraically, never as vectors, even though the resulting force is a vector.

NEET ESSENTIAL

* MUST REMEMBER

NEET PYQ PATTERN

Q. The minimum possible charge that can exist on an isolated body is

Ans. the elementary charge $e = 1.6 \times 10^{-19} \text{ C}$ – all observable charge is an integral multiple of e .

Conductors & Insulators

1. Conductors

Materials that allow charge to move freely through them, because they contain a large number of free electrons. Example: copper, silver, aluminium, the human body.

2. Insulators (Dielectrics)

Materials in which charge cannot move freely – electrons are tightly bound to their atoms. Example: rubber, glass, plastic, dry wood.

Property	Conductor	Insulator
Free electrons	Abundant	Almost none
Charge flow	Easy	Practically impossible
Charge given at a point	Spreads over surface	Stays localised
Example	Copper, silver	Rubber, glass

3. Charge Distribution

On an isolated conductor, excess charge always resides on the outer surface, and distributes itself so the conductor becomes an equipotential body (studied in Part VIII).

⚠ COMMON TRAP

Semiconductors (like silicon) are neither perfect conductors nor perfect insulators – their conductivity lies in between and depends on temperature and doping.

● NEET PYQ PATTERN

Q. A charged rod is brought close to (but not touching) a neutral metal sphere on an insulating stand. The nature of induced charge on the near face of the sphere is

Ans. opposite in sign to the charge on the rod (induction).

Methods of Charging

1. Charging by Friction

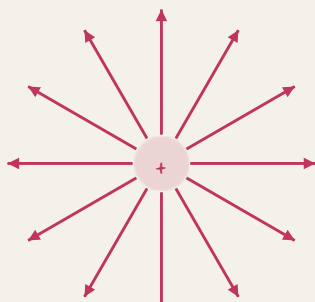
Rubbing two neutral bodies transfers electrons from one to the other. The body losing electrons becomes positive, the one gaining electrons becomes negative.

2. Charging by Conduction

A charged body is brought into direct contact with a neutral (or oppositely) conducting body. Charge flows until both reach the same potential. The neutral body gets charge of the same sign.

3. Charging by Induction

A charged body is brought near (without touching) a neutral conductor. Opposite charge is induced on the near face and like charge on the far face. Earthing the far face and then removing it leaves the conductor with a charge opposite to the inducing body.



● JEE FOCUS

Induction never changes the sign of charge on the inducing body, and the total charge of the (body + conductor) system is always conserved.

● NEET FOCUS

A charged balloon sticks to a neutral wall by induction, not conduction – the wall does not need to touch the balloon.

● NEET PYQ PATTERN

Q. An ebonite rod rubbed with fur becomes negatively charged. This happens because

Ans. electrons are transferred from the fur to the ebonite rod, leaving the fur positively charged.

Coulomb's Law

1. Statement

The electrostatic force of attraction or repulsion between two stationary point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them, acting along the line joining them.

$$F = k q_1 q_2 / r^2$$

SYMBOLS

F = electrostatic force (N) | q_1, q_2 = point charges (C) | r = distance between charges (m) | K = Coulomb constant

2. Coulomb Constant

$$k = 1 / (4\pi\epsilon_0) = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

ϵ_0 is the permittivity of free space. $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$.



● NEET PYQ PATTERN

Q. Two point charges placed 1 m apart in air exert a force F on each other. If the same charges are placed 1 m apart in a medium of dielectric constant $K = 4$, the new force is

Ans: $F/4$, since force is inversely proportional to the relative permittivity of the medium.

Vector Form & Superposition

1. Vector Form of Coulomb's Law

Force on charge q_1 due to q_2 , with $\hat{r}_{21}$ the unit vector from q_2 to q_1 :

$$\mathbf{F}_{12} = k \frac{q_1 q_2}{r^2} \cdot \hat{r}_{21}$$

SIGN RULE

If $q_1 q_2 > 0$ (like charges) $\rightarrow$ force is repulsive, along $+\hat{r}$. If $q_1 q_2 < 0$ (unlike charges) $\rightarrow$ force is attractive, along $-\hat{r}$.

2. Principle of Superposition

The net force on any charge due to several other charges is the vector sum of the forces exerted by each charge individually – as if the others were absent.

$$\mathbf{F}_{\text{net}} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots + \mathbf{F}_n \text{ (vector sum)}$$

● JEE FOCUS — TRICK

For 3 or more charges: resolve each force into x and y components before adding. Never add magnitudes directly unless forces are collinear.

● NEET PYQ PATTERN

Q. Three equal charges q are placed at the three corners of an equilateral triangle. The resultant force on any one charge acts

Ans: along the line joining that charge to the midpoint of the opposite side, directed away from the triangle (by symmetry of superposition).

PART II • COULOMB'S LAW

Medium & Permittivity

1. Effect of Medium

When charges are placed in a medium (instead of vacuum), the force between them decreases, because the medium gets polarised and partly opposes the field.

$$F_{\text{medium}} = k q_1 q_2 / (K r^2) = q_1 q_2 / (4\pi\epsilon_0 K r^2)$$

2. Relative Permittivity (Dielectric Constant)

$$K = \epsilon_r = \epsilon_{\text{medium}} / \epsilon_0 = F_{\text{vacuum}} / F_{\text{medium}}$$

K is a dimensionless number, $K = 1$ for vacuum/air, and $K > 1$ for all other dielectric media.

3. Coulomb's Law vs Newton's Law of Gravitation

Property	Coulomb's Law	Gravitation
Formula	$F = kq_1q_2/r^2$	$F = Gm_1m_2/r^2$
Nature	Attractive or repulsive	Always attractive
Depends on medium	Yes	No
Relative strength	$\sim 10^{36}$ times stronger	Very weak

● JEE ADVANCED

Both follow an inverse-square law and act along the line joining the two point objects.

● NEET PYQ PATTERN

Q. The ratio of electrostatic force to gravitational force between an electron and a proton in a hydrogen atom is of the order of
Ans: approximately 10^{39} – confirming that the electrostatic force dominates completely at atomic scales.

Coulomb's Law — Practice

Worked Example

Two point charges $+2 \mu\text{C}$ and $+3 \mu\text{C}$ are placed 30 cm apart in air. Find the force between them.

SOLUTION

$$F = K \frac{q_1 q_2}{r^2} = (9 \times 10^9 \times 2 \times 10^{-6} \times 3 \times 10^{-6}) / (0.3)^2$$

$$F = (9 \times 10^9 \times 6 \times 10^{-12}) / 0.09 = 0.054 / 0.09$$

$$F = 0.6 \text{ N (repulsive, since both charges are positive)}$$

NEET Quick Practice

- Two equal charges placed 1 m apart in air repel with force F . What is the force if the distance is halved?
- The force between two charges becomes $1/K$ times its original value when immersed in a medium. What does K represent?

NEET ESSENTIAL

* MUST REMEMBER

⚠ COMMON TRAP

Coulomb's law applies strictly to point charges. For extended/continuous bodies, break the charge into small elements dq and integrate.

Electric Field — Meaning

1. Why Electric Field?

Electric field is the region around a charge in which another charge experiences a force. It lets us describe the effect of a charge at every point in space, without needing a second charge to be actually present.

2. Test Charge

A small, positive charge q_0 used to probe the field at a point, small enough that it does not disturb the source charge distribution.

3. Definition & Formula

$$E = F / q_0$$

SYMBOLS

E = electric field intensity | F = force on test charge | q_0 = test charge (positive, small)

4. SI Unit & Dimension

Quantity	Unit	Dimension
Electric field (E)	N/C or V/m	$[M L T^{-3} A^{-1}]$

TRICK

Electric field is a vector. Its direction is the direction of force on a positive test charge.

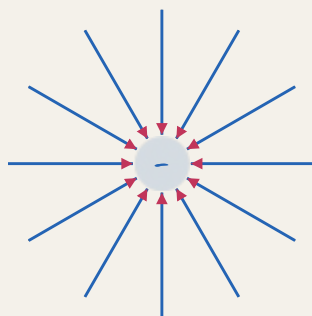
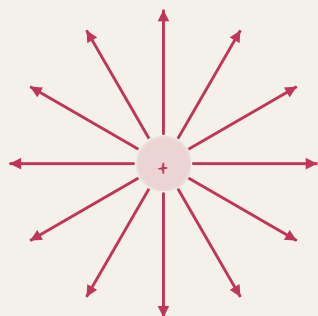
● NEET PYQ PATTERN

Q. A test charge is described as 'small' in the definition $E = F/q_0$ because
Ans. it must not disturb (redistribute) the source charges whose field is being measured.

Field Due to a Point Charge

1. Formula

$$E = k q / r^2 = q / (4\pi\epsilon_0 r^2)$$



2. Direction

- Points away from a positive charge (radially outward)
- Points towards a negative charge (radially inward)

3. Vector Form

$$\vec{E} = (1/4\pi\epsilon_0) (q/r^2) \hat{r}$$

⚠ COMMON TRAP

$E \propto 1/r^2$, not $1/r$. Doubling the distance reduces the field to one-fourth, not one-half.

● NEET PYQ PATTERN

Q. At a distance r from a point charge q , the electric field is E . At what distance will the field become $E/9$?

Ans. $3r$, since $E \propto 1/r^2$ and $(1/9) = 1/3^2$.

Field due to Multiple Charges

1. Superposition of Electric Field

The resultant field at a point due to several charges is the vector sum of fields due to each charge, treating the others as absent.

$$E_{\text{net}} = E_1 + E_2 + E_3 + \dots + E_n$$

2. Zero-Field Points

A point where the resultant electric field is zero. For two like charges, it lies between them (closer to the smaller charge); for two unlike charges, it lies outside, on the side of the smaller charge.



3. Using Symmetry

For symmetric charge arrangements (e.g. charges at corners of a square, or a ring of charge), components of field that are equal and opposite by symmetry can be eliminated before calculating – this saves significant time in JEE problems.

● JEE FOCUS

On the axis of a uniformly charged ring of radius R and charge Q , at distance x from the centre: $E = \frac{KQx}{(R^2 + x^2)^{3/2}}$. Maximum E occurs at $x = R/\sqrt{2}$.

● NEET PYQ PATTERN

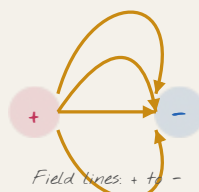
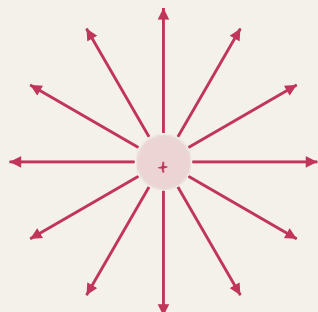
Q. Two point charges $+q$ and $+4q$ are placed a distance d apart. At what distance from $+q$ (on the line joining them) is the resultant electric field zero?

Ans: $d/3$ from $+q$ – set $Kq/x^2 = K(4q)/(d-x)^2$ and solve, giving $x = d/3$.

Electric Field Lines

1. Meaning

Electric field lines are imaginary curves such that the tangent at any point gives the direction of the electric field at that point.



2. Rules for Field Lines

- Start from positive charge (or infinity), end on negative charge (or infinity)
- Never intersect each other
- Denser lines $\rightarrow$ stronger field; sparse lines $\rightarrow$ weaker field
- Always perpendicular to a conducting surface
- Tend to contract lengthwise and repel sideways (explains attraction/repulsion)

WHY FIELD LINES NEVER INTERSECT

If two lines crossed, the field at that point would have two directions at once – impossible for a single-valued vector field.

● NEET PYQ PATTERN

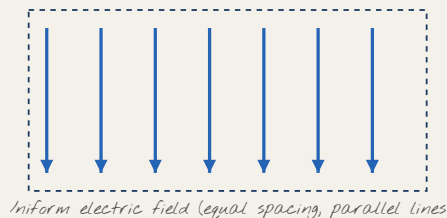
Q. Electric field lines due to a pair of equal and opposite charges are correctly represented as curves that

Ans. originate on the positive charge and terminate on the negative charge, never crossing each other.

Uniform Electric Field

1. Meaning

A field in which magnitude and direction of E are the same at every point in the region. Represented by equally spaced, parallel, straight field lines.



Uniform electric field (equal spacing, parallel lines)

2. Where it Occurs

- Between two oppositely charged parallel plates (away from edges)
- Far from a large uniformly charged plane sheet

3. Force & Motion in Uniform Field

$$F = qE \text{ (constant force } \Rightarrow \text{ uniform acceleration } a = qE/m)$$

● NEET FOCUS

A charged particle entering a uniform field perpendicular to it follows a parabolic path – exactly like projectile motion under gravity.

● NEET PYQ PATTERN

Q. An electron enters a uniform electric field E perpendicular to its initial velocity. Its path inside the field is

Ans. a parabola – exactly analogous to projectile motion, since it experiences constant force qE perpendicular to its initial velocity.

Continuous Charge Distribution

1. Why Continuous Distribution?

For rods, rings, sheets and spheres, charge is spread continuously, not as isolated points. We divide the body into infinitesimal elements of charge dq , apply Coulomb's law to each, and integrate.

2. Types of Charge Density

Density	Symbol	Definition	Unit
Linear	λ	charge / length	C/m
Surface	σ	charge / area	C/m ²
Volume	ρ	charge / volume	C/m ³

3. Setting Up the Integral

$$dE = k dq / r^2, \quad E = \int dE$$

TRICK

Always exploit symmetry first – perpendicular components of dE from symmetric elements cancel, leaving only the component along the axis of symmetry to integrate.

● NEET PYQ PATTERN

Q. For a uniformly charged rod, the correct expression for the linear charge density is

Ans: $\lambda = (\text{total charge } Q) / (\text{length } L)$, used to define $dq = \lambda dx$ for integration.

Field Due to a Charged Rod

Field on the Axis of a Finite Rod

For a rod of length L , linear charge density λ , at a point on its axis at distance a from the near end:

$$E = k\lambda L / [a(a+L)]$$

Field due to an Infinite Line Charge (preview)

As $L \rightarrow \infty$, this reduces to the elegant result derived later using Gauss's law (Part V):

$$E = \lambda / (2\pi\epsilon_0 r)$$

Notice: field due to a point charge falls as $1/r^2$, but field due to an infinite line falls only as $1/r$ – the geometry of the source changes how fast the field weakens.

⚠ COMMON TRAP

This $1/r$ result applies only near an effectively infinite line (or close to a long finite rod). Far from a finite rod, it behaves like a point charge ($1/r^2$).

● NEET PYQ PATTERN

Q. The electric field due to an infinitely long straight uniformly charged wire is proportional to

Ans: $1/r$, where r is the perpendicular distance from the wire – a direct result of Gauss's law with cylindrical symmetry.

Electric Field — Practice

Worked Example

A point charge of $+4 \mu\text{C}$ produces an electric field. Find its magnitude at a distance of 2 m in vacuum.

SOLUTION

$$E = Kq/r^2 = (9 \times 10^9 \times 4 \times 10^{-6}) / (2)^2$$

$$E = (36 \times 10^3) / 4 = 9 \times 10^3 \text{ N/C, directed away from the charge.}$$

NEET / JEE Quick Practice

- Two equal and opposite charges are placed at the vertices of an equilateral triangle. Find the resultant field at the third vertex (direction only).
- Sketch the field lines around a dipole and mark where the field is strongest.

NEET ESSENTIAL

JEE MAIN

More Practice

- Two point charges of $+9 \mu\text{C}$ and $+4 \mu\text{C}$ are placed 30 cm apart. Find the point on the line joining them where the field is zero.
- An electron is released from rest in a uniform field of $5 \times 10^4 \text{ N/C}$. Find its acceleration. ($m_e = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$)

AT A GLANCE — PART III

$$E = F/q_0 \quad | \quad E_{\text{point}} = Kq/r^2 \quad | \quad E_{\text{net}} = \Sigma E \text{ (vector sum)} \quad | \quad E_{\text{line(finite)}} = K\lambda L/[a(a+L)]$$

Electric Flux

1. Physical Meaning

Electric flux is a measure of the number of electric field lines passing through a given surface. It tells us how much of the field "flows" through an area.

2. Area Vector

A surface element is assigned a vector dA whose magnitude equals the area and whose direction is normal (perpendicular) to the surface.

3. Flux through a Plane Surface

$$\Phi = E \cdot A = EA \cos \theta$$

SYMBOLS

Φ = electric flux | E = electric field | A = area | θ = angle between E and the area vector (normal)

4. SI Unit

Quantity	Unit
Electric Flux (Φ)	$\text{N m}^2 \text{C}^{-1}$ (or $\text{V}\cdot\text{m}$)

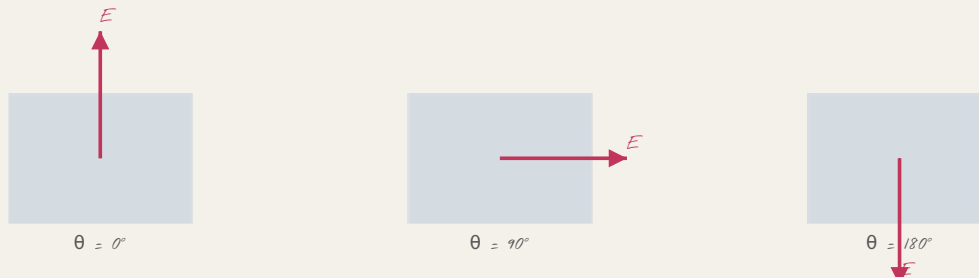
● NEET PYQ PATTERN

Q. The SI unit of electric flux can also be expressed as
Ans. volt-metre ($\text{V}\cdot\text{m}$), since $\Phi = E \cdot A$ and E has units of V/m .

PART IV • ELECTRIC FLUX

Flux — Angle Dependence

Flux at Different Angles



Angle θ	$\cos \theta$	Flux Φ	Meaning
0°	1	Maximum (+EA)	E parallel to normal (lines pass straight through)
90°	0	Zero	E parallel to the surface (no lines cross it)
180°	-1	Maximum negative (-EA)	E antiparallel to normal (lines enter, not exit)

Sign of Flux

- Flux is *positive* when field lines leave the surface (outward)
- Flux is *negative* when field lines enter the surface (inward)

⚠ COMMON TRAP

Flux depends only on the component of E along the normal, not on the field lying in the plane of the surface.

● NEET PYQ PATTERN

Q. The electric flux through a closed surface placed entirely inside a region of uniform electric field (with no charge enclosed) is

Ans: zero – every field line that enters the surface also exits it.

Flux — Non-Uniform Field & Practice

Non-Uniform Field / Curved Surface

When E varies over the surface, we divide it into small elements dA , compute the flux through each, and integrate.

$$\Phi = \int \mathbf{E} \cdot d\mathbf{A}$$

Worked Example

A uniform field $E = 200 \text{ N/C}$ passes through a flat surface of area 0.5 m^2 , making an angle of 60° with the surface normal. Find the flux.

SOLUTION

$$\Phi = EA \cos \theta = 200 \times 0.5 \times \cos 60^\circ$$

$$\Phi = 200 \times 0.5 \times 0.5 = 50 \text{ N m}^2/\text{C}$$

● JEE FOCUS — TRICK

Net flux through a closed surface in a uniform field is always zero — every line that enters also exits. This directly sets up Gauss's Law, next.

Gauss's Law

1. Why Gauss's Law?

For highly symmetric charge distributions (spheres, cylinders, planes), direct integration of Coulomb's law is long and difficult. Gauss's law gives the same field in just a few steps by using symmetry.

2. Key Terms

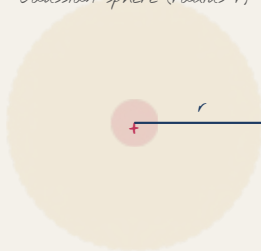
- **Closed surface** – a surface that completely encloses a volume
- **Gaussian surface** – an imaginary closed surface chosen to match the symmetry of the charge
- **Enclosed charge** – the net charge inside the Gaussian surface

3. Statement

The total electric flux through any closed surface equals $1/\epsilon_0$ times the net charge enclosed by that surface.

$$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enclosed}} / \epsilon_0$$

Gaussian sphere (radius r)



● NEET PYQ PATTERN

Q. Gauss's law is most useful in calculating electric fields when the charge distribution possesses

Ans. a high degree of symmetry – spherical, cylindrical, or planar – so a suitable Gaussian surface can be chosen.

Understanding Gauss's Law

1. Physical Interpretation

- Flux through the Gaussian surface depends only on the charge enclosed – charges outside contribute zero net flux
- The shape, size, and exact position of the surface do not matter, only which charges lie inside
- Gauss's law is always true, but is only useful for calculation when the charge distribution is symmetric

2. Gauss's Law vs Coulomb's Law

Coulomb's Law	Gauss's Law
Basic law for point charges	Derived from Coulomb's law + superposition
Direct integration needed for extended bodies	Uses symmetry, avoids integration
Always applicable	Easy to use only for symmetric charge distributions

⚠ COMMON TRAP

Q_{enclosed} means only the charge inside the Gaussian surface. Charges outside affect the field at individual points but contribute zero to the total flux.

● NEET PYQ PATTERN

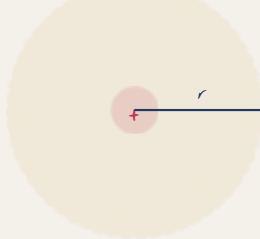
Q. If the net electric flux through a closed surface is zero, it can be concluded that

Ans: the net charge enclosed by the surface is zero – though individual point charges of both signs may still be present inside.

Gaussian Surface & Symmetry

Choosing the Right Gaussian Surface

Gaussian sphere (radius r)

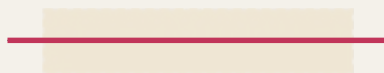


Gaussian cylinder around line charge (λ)



Symmetry	Gaussian Surface	Used For
Spherical	Concentric sphere	Point charge, charged sphere/shell
Cylindrical	Coaxial cylinder	Infinite line charge, charged cylinder
Planar	Pillbox (small cylinder/box)	Infinite plane sheet

Gaussian pillbox on infinite sheet (σ)



TRICK

Always choose a Gaussian surface on which E is either constant in magnitude, or zero, or perpendicular/parallel to dA everywhere – this is what makes the integral trivial.

● NEET PYQ PATTERN

Q. For calculating the field due to an infinite plane sheet of charge, the most suitable Gaussian surface is a

Ans: cylindrical pillbox with its flat faces parallel to the sheet.

Application: Infinite Line & Sheet

1. Infinite Line Charge

Linear charge density λ . Choose a coaxial cylindrical Gaussian surface of radius r and length L . E is radial and constant over the curved surface, flux through the flat ends is zero.

DERIVATION

$$E(2\pi rL) = \lambda L / \epsilon_0$$

$$E = \lambda / (2\pi\epsilon_0 r) \quad (\propto 1/r)$$

2. Infinite Plane Sheet of Charge

Surface charge density σ . Choose a pillbox Gaussian surface piercing the sheet, with flat faces of area A parallel to the sheet.

DERIVATION

$$E \cdot A + E \cdot A = \sigma A / \epsilon_0$$

$$E = \sigma / (2\epsilon_0) \quad (\text{uniform, independent of distance!})$$

★ MUST REMEMBER

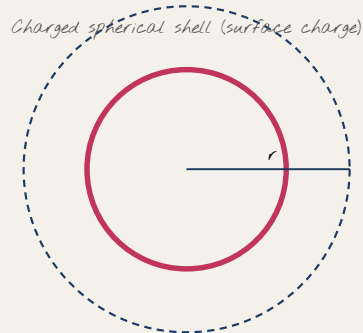
Between two oppositely charged parallel plates (each with surface density σ), the fields add up: $E = \sigma/\epsilon_0$. Outside the plates, the fields cancel: $E = 0$.

● NEET PYQ PATTERN

Q. Two large parallel plates carry surface charge densities $+\sigma$ and $-\sigma$. The electric field in the region between the plates is
Ans: σ/ϵ_0 directed from the positive towards the negative plate.

Application: Charged Spherical Shell

Thin Spherical Shell, Charge Q , Radius R



Region	Enclosed Charge	Electric Field
Inside ($r < R$)	Zero	$E = 0$
On surface ($r = R$)	Q	$E = kQ/R^2 = \sigma/\epsilon_0$
Outside ($r > R$)	Q	$E = kQ/r^2$

● NEET FOCUS

Outside the shell, it behaves exactly like a point charge Q placed at the centre.
Inside, the field is exactly zero everywhere – not just at the centre.

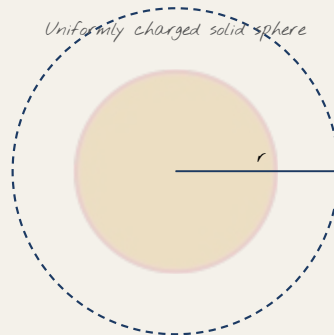
● NEET PYQ PATTERN

Q. A charge Q is placed at the centre of a spherical conducting shell of inner radius R_1 and outer radius R_2 . The charge induced on the inner surface of the shell is

Ans: $-Q$, and on the outer surface is $+Q$, by Gauss's law applied just inside the conductor (where $E = 0$).

Application: Uniformly Charged Solid Sphere

Non-Conducting Solid Sphere, Charge Q , Radius R



Region	Enclosed Charge	Electric Field
Inside ($r < R$)	$Q(r/R)^3$	$E = kQr / R^3 \ (\propto r)$
On surface ($r = R$)	Q	$E = kQ/R^2$
Outside ($r > R$)	Q	$E = kQ/r^2$

● JEE FOCUS — DERIVATION (INSIDE)

Charge enclosed at radius r : $q = Q(r/R)^3$ (uniform ρ)
 $E(4\pi r^2) = q/\epsilon_0 \Rightarrow E = kQr/R^3$, increasing linearly with r .

● NEET PYQ PATTERN

Q. For a uniformly charged non-conducting solid sphere of radius R , the electric field at $r = R/2$ (compared to the field at the surface) is
 Ans. half the surface field, since $E \propto r$ for $r \leq R$, so $E(R/2) = (1/2)E(R)$.

Conducting vs Non-Conducting Sphere

Key Distinction

Property	Conducting Sphere	Non-Conducting Sphere
Charge location	Entirely on surface	Uniformly through volume
Field inside ($r < R$)	$E = 0$	$E = kQr/R^3$ (increases with r)
Field at surface	$E = kQ/R^2$	$E = kQ/R^2$
Field outside	$E = kQ/r^2$	$E = kQ/r^2$



★ MUST REMEMBER

Outside any spherically symmetric charge distribution, the field looks exactly like that of a point charge at the centre – conducting or not.

⚠ COMMON TRAP

Students often assume E is zero inside every sphere. This is true only for a conductor or a hollow shell – not for a uniformly charged solid (non-conducting) sphere.

● NEET PYQ PATTERN

Q. Assertion: The electric field inside a uniformly charged solid sphere increases with distance from the centre. Reason: Enclosed charge increases with r inside the sphere.

Ans. Both Assertion and Reason are true, and Reason correctly explains the Assertion.

Electric Potential

1. Why Potential is Introduced

Electric field is a vector – often inconvenient to add. Electric potential is a scalar built from the work done by the field, making many problems far easier to solve.

2. Work & Potential Energy

Work done by an external agent in moving a charge (without changing its kinetic energy) against the electric field is stored as electrostatic potential energy.

3. Definition of Potential at a Point

$$V = W / q$$

SYMBOLS

V = electric potential | W = work done in bringing unit positive charge from infinity to that point | q = test charge

4. Nature & Unit

- Electric potential is a scalar quantity
- SI unit: volt (V), where $1 \text{ V} = 1 \text{ J/C}$

● NEET PYQ PATTERN

Q. Work done in moving a charge on an equipotential surface is always
Ans. zero, since there is no potential difference between any two points on the surface.

Potential Due to a Point Charge

Formula

$$V = (1/4\pi\epsilon_0) (Q/r) = kQ/r$$

Sign of Potential

- V is positive near a positive charge, and decreases to zero at infinity
- V is negative near a negative charge, and increases (towards zero) at infinity
- Unlike E , potential due to a point charge is a scalar – no direction, only sign

Reference Point

Potential is always defined relative to a reference. By convention, potential at infinity is taken as zero.

⚠ COMMON TRAP

V due to a point charge falls as $1/r$, while E falls as $1/r^2$ – do not confuse the two dependencies.

● NEET PYQ PATTERN

Q. The potential due to a point charge at a distance r is V . At what distance will the potential become $V/2$?

Ans: $2r$, since $V \propto 1/r$.

Potential Due to Multiple Charges

Superposition of Potential

Because potential is a scalar, the net potential at a point due to several charges is simply the algebraic sum (not vector sum) of the individual potentials.

$$V = V_1 + V_2 + V_3 + \dots + V_n$$

Potential is Scalar — Why It Matters

Electric Field (E)	Electric Potential (V)
Vector quantity	Scalar quantity
Added using vector rules (components)	Added by simple algebraic sum
Depends on direction from charge	Depends only on distance and sign of charge
Unit: N/C or V/m	Unit: volt (V)

● JEE FOCUS — TRICK

Potential at the centre of a system of charges may be non-zero even when the field there is zero (e.g. two equal charges at the ends of a diameter) — and vice-versa.

● NEET PYQ PATTERN

Q. Two equal and opposite charges are placed at the two ends of a diameter of a circle. The potential at the centre of the circle is
 Ans: zero, since the two equal-magnitude contributions (positive and negative) cancel algebraically.

Potential & Electric Field

1. Relation Between E and V

$$E = - dV/dr$$

The electric field at a point equals the negative of the potential gradient – the rate at which potential decreases with distance, in the direction of the field.

2. Potential Difference

$$V_B - V_A = - \int E \cdot dr \text{ (from A to B)}$$

3. Meaning of the Negative Sign

The field always points in the direction of *decreasing* potential – that is why moving along the field, potential falls; the minus sign captures this fact.

In a uniform field between parallel plates separated by distance d with potential difference V : $E = V/d$.

● NEET PYQ PATTERN

Q. If the electric potential in a region is constant, the electric field in that region must be

Ans: zero everywhere, since $E = -dV/dr$ and $dV/dr = 0$ for constant V .

Equipotential Surfaces

1. Definition

A surface on which the electric potential has the same value at every point.

2. Key Properties

- No work is done in moving a charge along an equipotential surface (since $V_B - V_A = 0$)
- Electric field is always perpendicular to an equipotential surface
- Two equipotential surfaces can never intersect
- Surfaces are closely spaced where the field is strong, widely spaced where it is weak

3. Shapes for Common Charge Systems

Charge System	Equipotential Surface
Point charge	Concentric spheres
Uniform field	Parallel planes perpendicular to E
Conductor (electrostatics)	The entire conductor surface is equipotential

★ MUST REMEMBER

$E \perp$ equipotential surface – this single fact settles most conceptual NEET/JEE questions on this topic.

● NEET PYQ PATTERN

Q. The equipotential surfaces due to an isolated point charge are

Ans. concentric spheres centred on the charge, with E everywhere perpendicular (radial) to them.

Potential of a Conducting Sphere

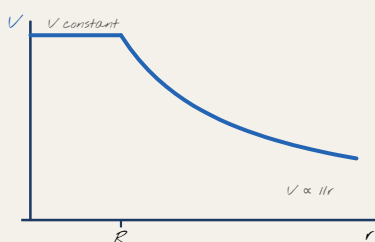
1. Important Points

- Inside a conductor, electric field is zero
- Hence, potential is constant throughout the conductor
- For a spherical conductor, potential is the same at all points on the surface
- Potential just outside the surface equals the surface potential

2. Potential due to a Charged Conducting Sphere

$$V = kQ/r \quad (r \geq R)$$

$$V = kQ/R \quad (r \leq R, \text{ constant})$$



NOTE

Potential is continuous at the surface ($r = R$), even though the field is discontinuous there.

● NEET PYQ PATTERN

Q. A charged conducting sphere of radius R has surface potential V . The potential at its centre is

Ans: also V – equal to the surface potential, since the interior is a field-free equipotential region.

Potential of a Non-Conducting Sphere

Uniformly Charged Solid Sphere

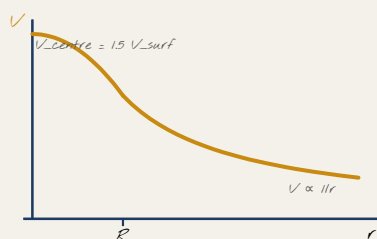
- Charge is uniformly distributed throughout the volume
- Electric field inside is not zero
- Potential inside varies with distance r

$$V = kQ/r \quad (r \geq R)$$

$$V = (kQ/2R)[3 - r^2/R^2] \quad (r \leq R)$$

At the Centre ($r = 0$)

$$V_0 = (kQ/2R) \times 3 = 3kQ/2R = (3/2) \times V_{\text{surface}}$$



NOTE

Potential at the centre is the maximum, and is exactly 1.5 times the surface potential.

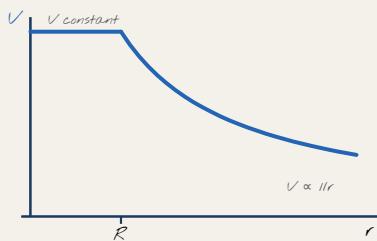
● NEET PYQ PATTERN

Q. For a uniformly charged non-conducting sphere, the ratio of potential at the centre to potential at the surface is

Ans: 3 : 2, i.e. $V_{\text{centre}} = (3/2) V_{\text{surface}}$

Potential — Important Results & Practice

Summary Graphs



Worked Example

Find the potential at a point 0.9 m from a point charge of $+3 \mu\text{C}$.

SOLUTION

$$V = KQ/r = (9 \times 10^9 \times 3 \times 10^{-6}) / 0.9$$

$$V = (27 \times 10^3) / 0.9 = 3 \times 10^4 \text{ V}$$

NEET ESSENTIAL

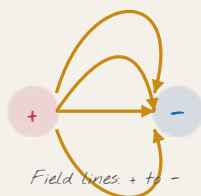
JEE MAIN

MUST REMEMBER

Electric Dipole

1. Definition

A pair of equal and opposite point charges ($\pm q$) separated by a small distance $2a$ is called an electric dipole.



2. Dipole Moment

$$p = q \times 2a$$

SYMBOLS

p = dipole moment (vector) | direction: from $-q$ to $+q$ | SI unit: coulomb-metre ($C \cdot m$)

● NEET PYQ PATTERN

Q. An electric dipole consists of charges $+2 \mu C$ and $-2 \mu C$ separated by 3 cm. Its dipole moment is

Ans: $p = q \times 2a = 2 \times 10^{-6} \times 0.03 = 6 \times 10^{-8} C \cdot m$, directed from $-q$ to $+q$.

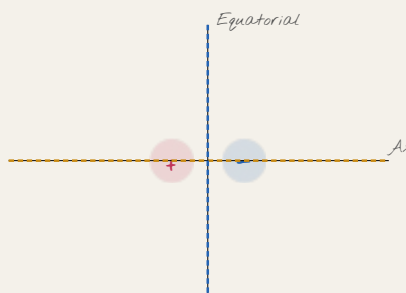
Potential & Field Due to a Dipole

1. Potential Due to a Dipole (general point)

$$V = (1/4\pi\epsilon_0) (p \cos \theta / r^2)$$

2. Electric Field Due to a Dipole (general point)

$$E = (1/4\pi\epsilon_0 r^3) \sqrt{3\cos^2 \theta + 1} \cdot p \text{ (magnitude)}$$



⚠ COMMON TRAP

These are far-field ($r \gg a$) approximate expressions. Exact expressions require the individual distances from each charge, especially in JEE Advanced-level problems.

● NEET PYQ PATTERN

Q. The electric potential due to a short dipole at a point on its equatorial line is
Ans: zero, since $\cos 90^\circ = 0$ in $V = Kp \cos\theta/r^2$.

Axial & Equatorial Positions

1. On the Axial Line ($\theta = 0^\circ$)

$$E_{\text{axial}} = (1/4\pi\epsilon_0) (2p / r^3)$$

2. On the Equatorial Line ($\theta = 90^\circ$)

$$E_{\text{equatorial}} = (1/4\pi\epsilon_0) (p / r^3)$$

★ MUST REMEMBER

$E_{\text{axial}} = 2 \times E_{\text{equatorial}}$ at the same distance r , and both point directions differ: axial field is along p , equatorial field is opposite to p .

3. Falloff with Distance

Unlike a point charge ($E \propto 1/r^2$), a dipole field falls off faster:

$$E_{\text{dipole}} \propto 1/r^3$$

● NEET FOCUS

This faster falloff happens because the fields of $+q$ and $-q$ nearly cancel at large distances, leaving only a small net effect.

● NEET PYQ PATTERN

Q. At the same distance r from a short dipole, the ratio of field magnitudes on the axial line to the equatorial line is

Ans: 2 : 1 ($E_{\text{axial}} = 2Kp/r^3$, $E_{\text{equatorial}} = Kp/r^3$).

Dipole in a Uniform Electric Field

1. Torque on a Dipole

A uniform field exerts equal and opposite forces on the two charges – net force is zero, but a net torque tends to align the dipole with the field.

$$\tau = \mathbf{p} \times \mathbf{E} = pE \sin \theta$$

2. Potential Energy of a Dipole

$$U = - \mathbf{p} \cdot \mathbf{E} = - pE \cos \theta$$

3. Equilibrium Positions

Orientation (θ)	Torque	Potential Energy	Equilibrium
$\theta = 0^\circ$ ($\mathbf{p} \parallel \mathbf{E}$)	Zero	$U = -pE$ (minimum)	Stable
$\theta = 180^\circ$ ($\mathbf{p}$ antiparallel $\mathbf{E}$)	Zero	$U = +pE$ (maximum)	Unstable
$\theta = 90^\circ$	Maximum (pE)	$U = 0$	—

● JEE FOCUS

Work done by an external agent rotating a dipole from θ_1 to θ_2 in a uniform field: $W = U(\theta_2) - U(\theta_1) = pE(\cos \theta_1 - \cos \theta_2)$.

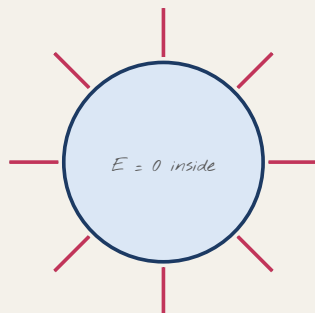
● NEET PYQ PATTERN

Q. An electric dipole is placed with its axis perpendicular to a uniform electric field. The torque on it is
Ans: maximum, equal to pE , since $\tau = pE \sin \theta$ is maximum at $\theta = 90^\circ$.

Conductors in Electrostatic Equilibrium

1. Electric Field Inside a Conductor

In electrostatic equilibrium, the electric field inside a conductor is always zero. If it were not, free electrons would keep moving – contradicting equilibrium.



2. Charge Distribution

Any excess charge on a conductor resides entirely on its outer surface. There is no charge in the bulk (interior) of the conductor.

3. Potential Inside a Conductor

Since $E = 0$ inside, the potential does not change from point to point – it is constant throughout the conductor, including its surface.

● NEET PYQ PATTERN

Q. A charge is placed inside a closed, uncharged conducting shell. The electric field just outside the shell depends on

Ans: the charge enclosed (and the geometry of the outer surface) – by Gauss's law, exactly as if all enclosed charge were at the centre for a spherical shell.

Conductor Surface Properties

1. Surface as an Equipotential

The entire surface of a conductor is an equipotential surface, and so is its interior – together they form a single equipotential body.

2. Field Just Outside the Surface

$$E = \sigma / \epsilon_0 \text{ (just outside a conductor)}$$

This field is always perpendicular to the conductor's surface (if it had a tangential component, charges would keep flowing along the surface).

3. Charge Concentration at Sharp Points

Surface charge density σ is higher where the radius of curvature is smaller – hence charge (and field) concentrates strongly at sharp points and edges. This is the working principle of a lightning conductor.

★ MUST REMEMBER

Smaller radius of curvature $\rightarrow$ larger $\sigma \rightarrow$ stronger local field. Sharp points can ionise surrounding air, allowing charge to leak or discharge.

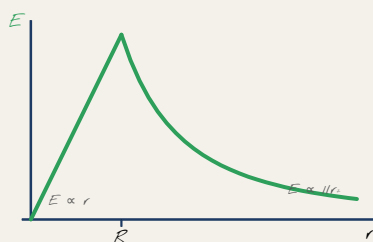
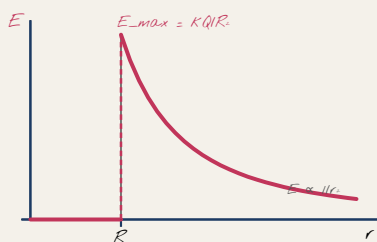
● NEET PYQ PATTERN

Q. The electric field is strongest near sharp points of a charged conductor because

Ans: surface charge density σ is highest where the radius of curvature is smallest, and $E = \sigma/\epsilon_0$ just outside the surface.

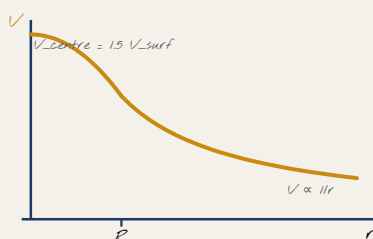
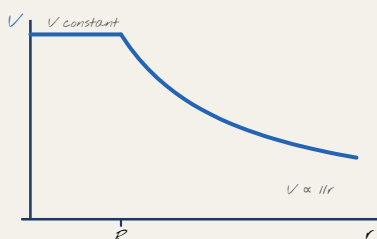
Important Graphs & Results

E vs r for Different Charge Distributions



Left: point charge / conducting sphere / shell ($E = 0$ inside, then $\propto 1/r^2$ outside) Right: uniformly charged solid sphere ($E \propto r$ inside, then $\propto 1/r^2$ outside)

V vs r for Different Charge Distributions



Left: conducting sphere/shell (V constant inside, then $\propto 1/r$ outside) Right: solid sphere (V maximum at centre, smooth fall inside, then $\propto 1/r$ outside)

KEY POINTS

- V is scalar, V is continuous everywhere (except right at a point charge, $r \rightarrow 0$)
- $E = -dV/dr$ (radial field), $V = -\int E \cdot dr$
- E is always perpendicular to an equipotential surface
- On any conductor, the surface is an equipotential surface

● NEET PYQ PATTERN

Q. For a uniformly charged solid sphere, the E - r graph shows E increasing linearly up to the surface and then

Ans. decreasing as $1/r^2$ outside the sphere, with the field continuous at $r = R$.

One-Page Electrostatics Revision

Concept	Formula / Result
Coulomb's Law	$F = kq_1q_2/r^2$
Electric field (point charge)	$E = kq/r^2$
Electric flux	$\Phi = EA \cos \theta$
Gauss's Law	$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{enc}/\epsilon_0$
Infinite line charge	$E = \lambda/2\pi\epsilon_0 r$
Infinite sheet charge	$E = \sigma/2\epsilon_0$
Just outside a conductor	$E = \sigma/\epsilon_0$
Potential (point charge)	$V = kq/r$
Field-potential relation	$E = -dV/dr$
Dipole moment	$\mathbf{p} = q \cdot 2\mathbf{a}$
Dipole field (axial / equatorial)	$2kp/r^3$ / kp/r^3
Torque, PE of dipole	$\tau = pE \sin \theta$, $U = -pE \cos \theta$

Sign Conventions & Common Mistakes

- Take potential at infinity as zero (reference point)
- E inside a conductor is always zero, inside a uniformly charged solid sphere it is not
- Flux depends only on enclosed charge, not on outside charges or surface shape
- Charge is scalar (add algebraically); field and force are vectors (add component-wise)

Exam-Day Strategy

- Read the question twice – note whether it asks for a conductor or a non-conductor before applying formulas
- For MCQs, check units first – a mismatched unit instantly eliminates 1-2 options
- For Assertion-Reason questions, verify the Assertion and Reason independently before checking if Reason explains Assertion
- Always sketch a quick diagram for dipole and Gauss's law problems – it prevents sign errors

Electrostatics Formula Bank

Charge & Coulomb's Law

$$Q = ne$$

$$F = kq_1q_2/r^2$$

$$k = 1/4\pi\epsilon_0$$

Electric Field & Flux

$$E = F/q_0$$

$$E = kq/r^2$$

$$\Phi = EA \cos\theta$$

Gauss's Law

$$\oint E \cdot dA = Q/\epsilon_0$$

$$E_{\text{line}} = \lambda/2\pi\epsilon_0 r$$

$$E_{\text{sheet}} = \sigma/2\epsilon_0$$

Electric Potential

$$V = W/q$$

$$V = kQ/r$$

$$E = -dV/dr$$

Dipole & Conductor

$$p = q \cdot 2a$$

$$\tau = pE \sin\theta$$

$$U = -pE \cos\theta$$

$$E_{\text{axial}} = 2kp/r^3$$

$$E_{\text{eq}} = kp/r^3$$

$$E_{\text{surface}} = \sigma/\epsilon_0$$

Constants Worth Memorising

$$k = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ SI}$$

HOW TO USE THIS PAGE

Cover the formulas with your hand and try to recall each one from its label – this active-recall method is far more effective than passively re-reading before an exam.

NEET Quick Practice

1. Two point charges separated by distance r exert force F on each other. What is the new force if the distance is tripled?
2. State whether electric field inside a hollow charged conducting sphere is zero or non-zero.
3. A charge of $6\ \mu\text{C}$ experiences a force of $0.03\ \text{N}$ at a point. Find the electric field at that point.
4. What is the SI unit of electric flux?
5. Assertion: Electric field lines never cross each other. Reason: A field can have only one direction at a given point. Check validity of Assertion and Reason.
6. Two charges $+q$ and $-q$ are separated by distance $2a$. Write the formula for the electric field on the equatorial line at distance r .
7. A conducting sphere of radius R carries charge Q . Find the potential at its centre.
8. Why is the potential same at every point on the surface of a charged conductor?
9. An electric dipole is placed in a uniform field with p parallel to E . What is the net force and net torque on it?
10. State Gauss's Law and mention the Gaussian surface used for an infinite plane sheet of charge.

Questions are original and framed for concept revision. For full worked solutions, review the relevant Part of this book.

EXAM STRATEGY

Attempt the direct-formula questions (1, 3, 4, 7) first to lock in easy marks, then move to the conceptual assertion-reason questions (2, 5, 8, 9) which need a little more thought.

JEE Concept Builder

1. Q1. Three equal charges q are placed at the vertices of an equilateral triangle of side a . Find the net force on any one charge.
2. Q2. A ring of radius R carries a uniformly distributed charge Q . Find the distance from the centre, along the axis, at which the electric field is maximum.
3. Q3. A solid non-conducting sphere of radius R has uniform volume charge density ρ . Find the electric field at $r = R/2$ from the centre.
4. Q4. Using Gauss's law, derive the electric field at a point outside an infinitely long charged cylindrical shell of radius R and linear charge density λ .
5. Q5. An electric dipole of moment p is placed at the origin along the x -axis. Find the ratio of electric field magnitudes at equal distances r on the axial and equatorial lines.
6. Q6. Sketch the V - r graph for a uniformly charged non-conducting sphere and identify the region where the slope of the graph equals $-E$.
7. Q7. Two identical conducting spheres with charges Q and $-Q/3$ are brought into contact and separated. Find the new charge on each sphere.
8. Q8. A short dipole of moment p is placed at the centre of a sphere of radius r . Find the total electric flux through the sphere.
9. Q9. Two point charges $+q$ and $+4q$ are fixed 90 cm apart. Find the position of a third charge Q so that the entire system is in equilibrium.

COMPACT ANSWER KEY (FINAL NUMERICAL/RELATIONAL RESULTS)

1. $F_{\text{net}} = \sqrt{3} K q^2 / a^2$ 2. $x = R/\sqrt{2}$ 3. $E = K \rho R / \epsilon_0$ ($\propto r$, use $r=R/2$) 4. $E = \lambda / 2\pi \epsilon_0 r$ 5. $E_{\text{axial}} : E_{\text{eq}} = 2 : 1$ 6. Slope = $-E$ at every r 7. Each sphere gets $Q/3$ 8. Flux = 0 (net charge of a dipole is zero) 9. Q at 30 cm from $+q$ charge = $-4q/9$ (equilibrium is unstable)

JEE TIP

In multi-charge equilibrium problems, first fix positions using force balance, then check the sign and magnitude of the third charge separately – the two conditions are independent.

ELECTROSTATICS

MASTER REVISION

Remember

- Charge creates electric field.
- Electric field produces force on charge.
- Electric potential is scalar.
- Electric field is related to the potential gradient.
- Gauss's law connects electric flux with the charge it encloses.



Thank You for Studying with VS Sir

You've completed Electrostatics – the foundation of Current Electricity, Capacitors & Magnetism.
Keep this book close during revision week.

📞 Doubts? WhatsApp +91 96024 05311

"Understand the Physics. Visualize the Concept. Master the Problem."

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Concepts + Practice = Success